ANNEX II

Cambodia: Debt Sustainability Analysis[9]

External debt—of which almost 60 percent is owed to the United States and the Russian Federation—constitutes over 90 percent of Cambodia's public debt. At end-2005, Cambodia's total external public debt was about $3.2 billion (50½ percent of GDP), while domestic debt amounted to 3½ percent of GDP, denominated entirely in local currency. Considerable progress has been made in recent months in resolving the issue of outstanding debt obligations to the United States and the Russian Federation.

This joint WB/IMF debt sustainability assessment comes to the conclusion that Cambodia's debt is on a sustainable path, but the current risk of debt distress remains high. If debt rescheduling agreements are reached, the risk of debt distress will moderate, though low revenue collections and extreme external shocks will continue to pose risks to debt sustainability. The analysis is based on the assumption that real GDP growth will be sustained at 6 percent and exports will grow at around 9 percent a year, with economic activity underpinned by garment exports and tourism. Business environment reforms are expected to help improve competitiveness. Although oil production is expected during the projection period, the potential impact is not included in the baseline or alternative scenarios due to uncertainty as to its scale, timing, and fiscal return. Foreign aid is assumed to remain broadly stable in nominal U.S. dollar terms. On the fiscal front, fiscal consolidation is expected to begin with the 2006 budget, underpinned by improvements in the tax and customs administration. The primary balance will improve over the medium term with improved fiscal performance. New borrowing is projected to be highly concessional and external debt owed to United States and the Russian Federation is assumed to be restructured in 2006.

The assessment also incorporates Cambodia's receipt of debt relief from the Fund under the Multilateral Debt Relief Initiative (MDRI). As outlined in the published staff assessment (12/08/05), Cambodia satisfied the three conditions for eligibility, thus qualifying to receive immediate debt relief from the Fund of SDR57 million (US$82 million).[10] The impact of this measure on the NPV of external debt-to-GDP ratio is about 1¼ percentage points, with debt relief providing around 1¼ percent of additional revenue in 2006 that will be spent over the medium term on poverty reducing projects.

A. The Current Situation

Cambodia's two largest creditors are the Russian Federation and the United States. Since Cambodia is not servicing its debt to either creditor and is making efforts to conclude agreements with both creditors under the framework of the Paris Club, the DSA assumes a

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[9] This DSA has been prepared jointly with the World Bank staff based on a new framework considered by the Boards of the Fund and the Bank in April 2005.

[10] Three criteria are: (i) good macroeconomic performance, (ii) implementation of a poverty reduction policy, and (iii) adequate public expenditure management systems.

debt restructuring agreement with the United States and Russia in 2006. As a consequence, Cambodia's net present value (NPV) of external debt-to-GDP ratio would drop from 40½ percent at end-2005 to 22½ percent in 2006.

The outstanding differences between the United States and Cambodia, until recently, have involved the total amount of principal. After clarification of outstanding legal issues, the countries have agreed on a total obligation of US$162 million. The next steps required to settle this issue include signing a bilateral agreement based on the 1995 Paris Club agreement (i.e., flow rescheduling on Naples terms assuming a 40-year maturity, 16-year grace period, and an interest rate of 3 percent), and calculating arrears up to now. It is expected that Cambodia would then request a rescheduling of those arrears through the Paris Club. While the exact terms of such rescheduling are not certain, staff assumes that the Naples terms will be applied.

Cambodia and the Russian Federation have discussed rescheduling with a stock operation. They have agreed that (i) after applying the stipulated exchange rate and the 70 percent upfront discount, the total debt would be US$457 million, and (ii) interest rate for the pre-cut-off-date debt would be about 0.8 percent, with concessional repayment schedule. However, the countries have disagreed on the classification of US$40 million as either pre-cut-off-date or post-cut-off-date debt, and the interest rate on the post-cut-off-date debt. The current Fund staff assumption for the macroeconomic framework is (i) the disputed US$40 million is classified as post-cut-off-date debt, and (ii) the interest rate for the post-cut-off-date is set according to the Russian offer at the latest bilateral meeting.

Cambodia ranks as a 'weak performer' in terms of its policies and institutions according to the latest World Bank's Country Policy and Institutional Assessment (CPIA). Indicative debt-burden thresholds for countries in this category are a NPV of external debt-to-exports of 100 percent, to GDP of 30 percent and to revenues of 200 percent, and debt service ratios to exports and to revenues of 15 and 25 percent, respectively.

B. Public Debt Sustainability

Cambodia's public debt burden indicators reflect the country's weak fiscal revenue base. The NPV of public debt-to-revenue ratio, which was 348 percent at end-2005, will remain high at around 182 percent even under the assumption that the bilateral restructuring agreements with the United States and Russia will be concluded in 2006.[11] The ratio will remain below 200 percent in the medium term, with improved revenue performance as assumed in the baseline, declining to 161 percent by 2011. However, the debt-to-revenue threshold is breached for multiple years under all the bounds tests, indicating the risk that low revenues pose to debt sustainability. Under the baseline scenario, the stock of public debt, which amounted to 54 percent of GDP at the end of 2005, is projected to decline to 36½ percent in 2006, or around 25 percent in NPV terms. Given the assumption that the

[11] Due to the MDRI relief being booked as a one-off capital revenue item in the fiscal accounts, the ratios to revenue drop in 2006 and increase in 2007.

primary deficit improves gradually in the medium term and remains at about 1 percent of GDP in the long run, the NPV of public sector debt-to-GDP ratio would drop to 8½ percent by 2026.

Cambodia's debt sustainability hinges on the improvement of weak fiscal performance. A continuation of 2005's fiscal performance would lead to a greater decline in the NPV of public debt-to-GDP ratio than under the baseline (Test A2). However, maintenance of primary deficit at the 2004 level—at 3 percent of GDP—would result in the NPV of public debt-to-GDP ratio remaining above 30 percent and the NPV of debt-to-revenue ratio remaining above 200 percent in the scenario, both above their respective indicative thresholds. A one-time exchange rate depreciation (Test B4) would raise the NPV of debt-to-GDP ratio by 9 percentage points and the NPV of debt-to-revenue ratio by 70 percentage points in 2007 relative to the baseline.[12] The difference between this stress test and the baseline is projected to decrease only slowly over time. However, the debt-service-to-revenue ratio remains relatively low in most scenarios since it is assumed that concessionality of external debt is high in the medium term.

C. External Debt Sustainability

Cambodia's external debt burden indicators are likely to remain below their indicative thresholds following conclusion of rescheduling agreements. The baseline scenario, however, shows that the NPV of debt-to-revenue ratio will remain close to the threshold in the medium term.[13] The ratio to GDP of the NPV of external debt will remain below the 30 percent indicative threshold in the medium-term, although breaching the threshold under some stress tests. The NPV of debt-to-export ratio, however, is 32½ percent in 2006, well below its indicative debt burden threshold of 100 percent. The debt service-to-exports ratio remains between 1 and 1½ percent in the long term reflecting the fact that new borrowing is assumed to be highly concessional.

The ratio to GDP of the NPV of external debt would breach its indicative threshold in the event of an export shock, indicating that Cambodia's debt sustainability remains vulnerable to exogenous shocks.[14] Lower export growth would result in a NPV of debt-to-GDP ratio of more than the indicative threshold of 30 percent in 2008 and 2009, remaining above the threshold in the medium term. The NPV of debt-to-export ratio would also substantially deteriorate by 19 percentage points (relative to the baseline scenario) to 53 percent in 2008. However, the ratio remains below its indicative debt burden threshold. Debt service ratios would remain below their indicative thresholds throughout the entire projection period even under the most extreme stress test, since the new borrowing is

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[12] This test grossly overestimates the negative impact since Cambodia is highly dollarized and nearly all the transactions are executed in U.S. dollars.

[13] The temporary drop in the ratio in 2006 is due to the treatment of the MDRI relief as capital revenue.

[14] One standard deviation shock is very large (12 percent) since Cambodia's exports have expanded very rapidly for past several years. Cambodia has never experienced such a large negative shock since it started exporting garment products in mid-1990s.

assumed to be highly concessional. If no agreement is reached with the United States and the Russian Federation, the external debt-to-GDP ratio will remain at a high level. Consequently, in the absence of debt service to their largest creditors, Cambodia's debt service payments will remain low (Figure A-3).

D. Impact of Less-Concessional Loan

The authorities are planning to borrow on less concessional terms. The authorities are considering financing critical infrastructure investment projects through the Asian Development Bank's Ordinary Capital Resource (OCR) lending, which may have only limited concessionality.[15] Additional less concessional resources may be made available from China, which could add to the debt burden. While it is understandable that better infrastructure is needed to foster higher living standards and a favorable business environment, attention should be paid to its implication for future expenditure, given the country's low revenue base and weak institutions. Careful assessment is needed to plan for such large infrastructure investments, and transactions should be monitored closely and transparently. If these loans are made available to the private sector, with political risk guarantee being provided by the Government, the contingent liabilities implications need to be considered. The current staff simulation shows that if such less-concessional borrowing were limited to US$20 million per year, the revenue impact would be about ½ to ¾ percent in the medium-term (Figure A-3).

E. Conclusion

Staffs come to the conclusion that Cambodia's debt is sustainable, and that the current risk of debt distress is high. Once rescheduling agreements are concluded, the risk will moderate sharply thereafter if baseline assumptions—improved revenue performance and sustained growth—hold. The NPV of public sector debt-to-GDP ratio falls below the indicative threshold in the medium-term after rescheduling of United States and Russian debts, and the NPV of debt-to-revenue ratio remains near its indicative threshold of 200 percent in the medium-term under the baseline, and is highly sensitive to fluctuations in macroeconomic and fiscal parameters. While the NPV of external debt-to-GDP ratio is slightly below its indicative debt burden threshold in the medium-term, stress tests indicate that an extreme export shock could substantially increase Cambodia's probability of debt distress in the medium term. Continuation of prudent economic policies—including strengthening its revenue base, which currently stands at less than 11 percent of GDP, and thus improving its primary balance—and export diversification should ameliorate these concerns and allow debt dynamics to be sustainable. Further, the positive impact of oil production on exports and revenues in the longer term would obviously reduce the risks Cambodia faces. Close attention, however, needs to be paid to Cambodia's debt management strategy, especially regarding less concessional loans, as it may impair sustainability, especially in the presence of a low revenue base and weak institutions.

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[15] The interest rate of OCR is in line with market rate. The lending period is from 15-20 years and the grace period is 2-7 years.

Table A1. Cambodia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2003-2026
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2003	2004	2005	Historical Average 5/	Standard Deviation 5/	2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2026	2012-26 Average
Public sector debt 1/	64.3	61.1	54.0			36.5	36.3	35.7	34.9	33.8	32.9		26.5	13.6	
o/w foreign-currency denominated	59.8	56.6	50.6			33.9	34.3	34.2	33.8	33.1	32.3		25.9	12.9	
Change in public sector debt	2.2	-3.3	-7.1			-17.5	-0.2	-0.6	-0.8	-1.1	-0.9		-1.4	-1.7	
Identified debt-creating flows	0.2	-4.3	-7.2			-18.7	-0.7	-0.7	-1.0	-1.4	-1.5		-0.9	-0.1	
Primary deficit	3.5	2.8	1.0	2.3	1.0	0.8	2.2	1.8	1.4	1.0	0.7	1.3	1.0	1.0	1.0
Revenue and grants	12.0	12.2	12.6			13.8	12.9	13.2	13.6	13.8	14.1		13.9	13.9	
of which : grants	2.3	1.7	2.2			1.7	1.6	1.4	1.3	1.1	1.0		0.7	0.2	
Primary (noninterest) expenditure	15.5	14.9	13.7			14.6	15.1	15.1	15.0	14.7	14.9		14.9	14.9	
Automatic debt dynamics	-4.2	-7.9	-8.9			-3.5	-2.9	-2.5	-2.4	-2.3	-2.2		-1.8	-1.0	
Contribution from interest rate/growth differential	-5.7	-6.9	-8.4			-3.2	-2.5	-2.3	-2.2	-2.1	-2.0		-1.6	-0.9	
of which : contribution from average real interest rate	-0.8	-1.1	-1.2			-0.7	-0.2	-0.2	-0.2	-0.2	-0.2		-0.1	-0.1	
of which : contribution from real GDP growth	-4.9	-5.8	-7.2			-2.6	-2.2	-2.1	-2.0	-1.9	-1.9		-1.5	-0.8	
Contribution from real exchange rate depreciation	1.5	-1.0	-0.5			-0.3	-0.4	-0.2	-0.2	-0.2	-0.2		...	...	
Other identified debt-creating flows	0.9	0.8	0.6			-16.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.9	0.8	0.6			-16.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.0	1.1	0.2			1.2	0.5	0.1	0.1	0.3	0.6		-0.5	-1.6	
NPV of public sector debt	52.7	48.6	44.0			25.1	25.4	25.2	24.4	23.5	22.8		17.5	8.4	
o/w foreign-currency denominated	48.2	44.1	40.5			22.5	23.4	23.7	23.3	22.8	22.2		16.8	7.7	
o/w external	48.2	44.1	40.5			22.5	23.4	23.7	23.3	22.8	22.2		16.8	7.7	
NPV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.1	3.3	1.5			1.5	2.9	2.6	2.2	1.8	1.7		2.1	1.5	
NPV of public sector debt-to-revenue ratio (in percent) 3/	439.9	400.0	347.8			182.2	197.4	190.4	179.5	170.4	161.3		125.9	60.5	
o/w external	401.8	363.2	320.8			163.3	182.0	179.3	171.8	165.6	157.0		121.1	55.7	
Debt service-to-revenue ratio (in percent) 3/ 4/	4.6	4.4	3.7			4.8	5.7	5.9	6.1	6.4	6.6		7.9	4.3	
Primary deficit that stabilizes the debt-to-GDP ratio 6/	1.3	6.1	8.1			2.4	2.4	2.4	2.2	2.1	1.6		2.3	2.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	8.6	10.0	13.4	9.5	2.6	5.0	6.5	6.1	5.8	5.8	5.8	5.8	5.8	5.8	5.8
Average nominal interest rate on forex debt (in percent)	0.4	0.4	0.4	0.3	0.1	0.9	1.6	1.5	1.5	1.4	1.5	1.4	1.5	1.2	1.4
Average real interest rate on domestic currency debt (in percent)	...	...	-5.1	...	...	-5.2	-4.4	-3.6	-3.9	-4.2	-4.7	-4.3	2.9	2.9	2.9
Real exchange rate depreciation (in percent, + indicates depreciation)	2.8	-1.8	-1.1	1.3	3.8	-0.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.2	5.3	5.7	2.0	3.0	5.2	4.0	3.0	3.0	3.0	3.0	3.5	3.0	3.0	3.0
Growth of real primary spending (deflated by GDP deflator, in percent)	1.3	5.8	3.8	10.2	7.0	12.0	9.9	6.0	5.2	4.0	6.9	7.3	5.8	5.8	5.8
Grant element of new external borrowing (in percent)	...	...	48.4			47.6	47.5	47.6	48.9	49.0	49.0	48.3	49.0	49.6	...

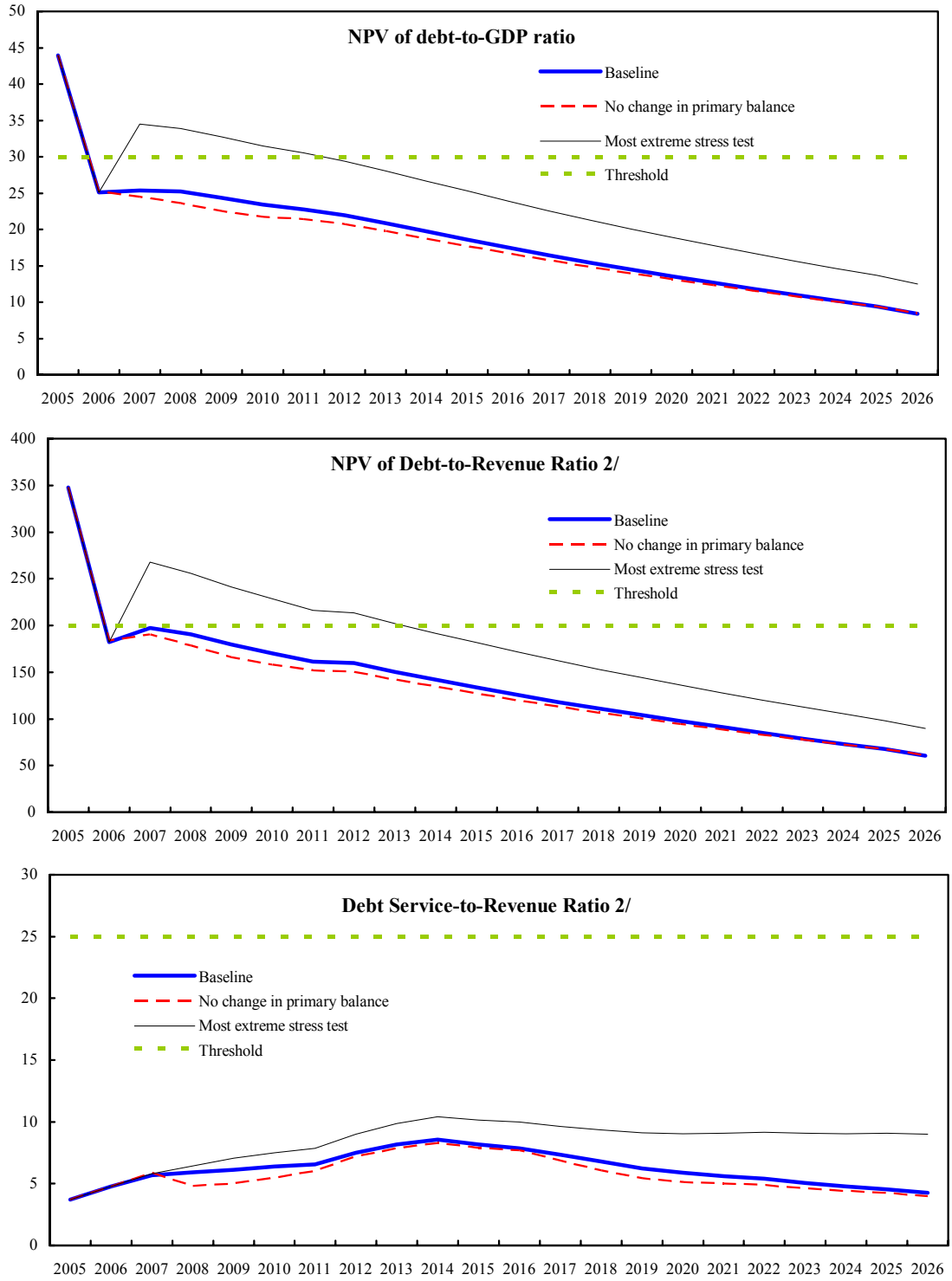
Sources: Country authorities; and Fund staff estimates and projections.
1/ General government gross debt is used.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues including grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 7 years, since the consistent balance of payments data are available only since 1998.
6/ Debt rescheduling with the Russian Federation and the United States is assumed in 2006.

Table A2. Cambodia: Sensitivity Analysis for Key Indicators of Public Debt 2006-2026

	Estimate	Projections						
	2006	2007	2008	2009	2010	2011	2016	2026
NPV of Debt-to-GDP Ratio								
Baseline	25	25	25	24	23	23	18	8
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	25	25	25	24	25	24	23
A2. Primary balance is unchanged from 2005	25	25	24	23	22	21	17	8
A3. Permanently lower GDP growth 1/	25	26	26	25	25	24	22	21
B. Bound tests								
B1. Real GDP growth is at baseline minus one standard deviations in 2007-2008	25	26	27	27	26	26	23	17
B2. Primary balance is at baseline minus one standard deviations in 2007-2008	25	26	27	26	25	24	19	9
B3. Combination of B1-B2 using one half standard deviation shocks	25	26	27	26	25	24	18	9
B4. One-time 30 percent real depreciation in 2007	25	34	33	32	30	29	21	11
B5. 10 percent of GDP increase in other debt-creating flows in 2007	25	34	34	33	31	31	24	12
NPV of Debt-to-Revenue Ratio 2/								
Baseline	182	197	190	179	170	161	126	60
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	182	196	190	183	180	177	172	167
A2. Primary balance is unchanged from 2005	184	190	179	166	158	152	120	61
A3. Permanently lower GDP growth 1/	182	199	193	184	178	171	155	148
B. Bound tests								
B1. Real GDP growth is at baseline minus one standard deviations in 2007-2008	182	203	204	196	191	184	164	121
B2. Primary balance is at baseline minus one standard deviations in 2007-2008	182	204	203	191	182	172	135	66
B3. Combination of B1-B2 using one half standard deviation shocks	182	203	200	188	179	169	131	61
B4. One-time 30 percent real depreciation in 2007	182	267	252	234	219	205	154	78
B5. 10 percent of GDP increase in other debt-creating flows in 2007	182	268	256	241	229	216	172	90
Debt Service-to-Revenue Ratio 2/								
Baseline	4.8	5.7	5.9	6.1	6.4	6.6	7.9	4.3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	4.8	5.5	6.0	6.6	7.3	8.0	10.3	12.2
A2. Primary balance is unchanged from 2005	4.8	5.9	4.8	5.0	5.5	6.0	7.7	4.0
A3. Permanently lower GDP growth 1/	4.8	5.7	6.0	6.4	6.8	7.1	9.6	10.2
B. Bound tests								
B1. Real GDP growth is at baseline minus one standard deviations in 2007-2008	4.8	5.8	6.4	7.0	7.5	7.8	10.0	9.0
B2. Primary balance is at baseline minus one standard deviations in 2007-2008	4.8	5.7	6.8	7.4	7.0	7.1	8.3	5.0
B3. Combination of B1-B2 using one half standard deviation shocks	4.8	5.7	6.5	6.8	6.7	6.9	8.1	4.4
B4. One-time 30 percent real depreciation in 2007	4.8	6.5	7.6	7.9	8.1	8.3	9.5	5.9
B5. 10 percent of GDP increase in other debt-creating flows in 2007	4.8	5.7	15.9	9.7	9.2	9.1	10.0	8.0

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Figure A1. Cambodia: Indicators of Public Debt Under Alternative Scenarios, 2005-2026 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2016.
2/ Revenue including grants.

Table A3. Cambodia: External Debt Sustainability Framework, Baseline Scenario, 2003-2026 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Estimate	Projections								
	2003	2004	2005			2006	2007	2008	2009	2010	2011	2005-10 Average	2016	2026	2012-26 Average
External debt (nominal) 1/	59.8	56.6	50.6			33.9	34.3	34.2	33.8	33.1	32.3		25.9	12.9	
o/w public and publicly guaranteed (PPG)	59.8	56.6	50.6			33.9	34.3	34.2	33.8	33.1	32.3		25.9	12.9	
Change in external debt	1.6	-3.2	-6.0			-16.7	0.4	-0.1	-0.4	-0.7	-0.8		-1.4	-1.7	
Identified net debt-creating flows	-1.8	-7.6	-11.0			-2.6	-1.4	-2.4	-2.5	-3.8	-3.8		-5.3	-7.6	
Non-interest current account deficit	3.3	2.0	3.9	2.7	1.3	5.2	5.9	4.8	4.6	3.2	3.0		1.3	-1.3	0.5
Deficit in balance of goods and services	10.2	7.4	8.7			9.5	9.8	8.4	7.7	5.9	5.3		1.5	-3.1	
Exports	57.5	64.5	64.1			68.6	68.8	68.8	68.6	68.3	68.1		69.1	75.5	
Imports	67.6	71.9	72.8			78.1	78.7	77.2	76.3	74.2	73.4		70.7	72.5	
Net current transfers (negative = inflow)	-10.5	-9.4	-8.5	-10.5	1.3	-8.4	-8.2	-7.9	-7.5	-7.1	-6.6		-4.4	-1.8	-3.6
Other current account flows (negative = net inflow)	3.6	3.9	3.8			4.2	4.3	4.4	4.4	4.4	4.3		4.1	3.6	
Net FDI (negative = inflow)	-1.6	-2.3	-6.1	-3.9	1.8	-5.9	-5.8	-5.8	-5.7	-5.6	-5.5		-5.5	-5.7	-5.6
Endogenous debt dynamics 2/	-3.5	-7.3	-8.9			-2.0	-1.5	-1.5	-1.4	-1.4	-1.3		-1.1	-0.6	
Contribution from nominal interest rate	0.4	0.4	0.3			0.4	0.5	0.5	0.5	0.4	0.5		0.4	0.2	
Contribution from real GDP growth	-4.7	-5.2	-6.3			-2.4	-2.0	-1.9	-1.8	-1.8	-1.8		-1.5	-0.8	
Contribution from price and exchange rate changes	0.8	-2.5	-2.9			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	3.3	4.5	5.0			-14.1	1.8	2.3	2.1	3.1	3.0		3.9	5.9	
o/w exceptional financing 8/	0.9	0.8	0.6			-16.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	48.2	44.1	40.5			22.5	23.4	23.7	23.3	22.8	22.2		16.8	7.7	
In percent of exports	83.8	68.4	63.3			32.7	34.0	34.5	34.0	33.4	32.5		24.4	10.2	
NPV of PPG external debt	48.2	44.1	40.5			22.5	23.4	23.7	23.3	22.8	22.2		16.8	7.7	
In percent of exports	83.8	68.4	63.3			32.7	34.0	34.5	34.0	33.4	32.5		24.4	10.2	
Debt service-to-exports ratio (in percent) 7/	0.9	1.0	0.9			0.8	1.1	1.2	1.2	1.3	1.4		1.5	0.7	
PPG debt service-to-exports ratio (in percent) 7/	0.9	1.0	0.9			0.8	1.1	1.2	1.2	1.3	1.4		1.5	0.7	
Total gross financing need (billions of U.S. dollars)	0.1	0.0	-0.1			0.0	0.1	0.0	0.0	-0.1	-0.2		-0.5	-2.3	
Non-interest current account deficit that stabilizes debt ratio	1.8	5.1	9.9			21.9	5.5	4.9	5.0	3.9	3.8		2.7	0.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	8.6	10.0	13.4	9.5	2.6	5.0	6.5	6.1	5.8	5.8	5.8	5.8	5.8	5.8	5.8
GDP deflator in US dollar terms (change in percent)	-1.3	4.3	5.4	1.0	3.2	1.8	2.5	2.5	2.6	2.6	2.7	2.4	2.7	2.7	2.7
Effective interest rate (percent) 5/	0.7	0.7	0.7	0.7	0.0	0.8	1.6	1.5	1.5	1.4	1.5	1.4	1.5	1.2	1.4
Growth of exports of G&S (US dollar terms, in percent)	11.0	28.7	18.7	22.9	12.1	14.5	9.4	8.8	8.2	8.0	8.4	9.6	9.2	9.9	9.4
Growth of imports of G&S (US dollar terms, in percent)	13.4	22.0	20.9	18.9	9.1	14.7	9.9	6.8	7.3	5.5	7.5	8.6	8.3	9.4	8.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	47	47	47	48	49	49	48	49	-1	46
Memorandum item:															
Nominal GDP (billions of US dollars)	4.6	5.3	6.3			6.7	7.3	8.0	8.7	9.4	10.2		15.5	35.5	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 7 years, since the consistent balance of payments data are only available since 1998.
7/ Debt services are cash basis until 2005.
8/ Includes effect of MDRI debt relief. Debt rescheduling with the Russian Federation and the United States is also assumed in 2006.

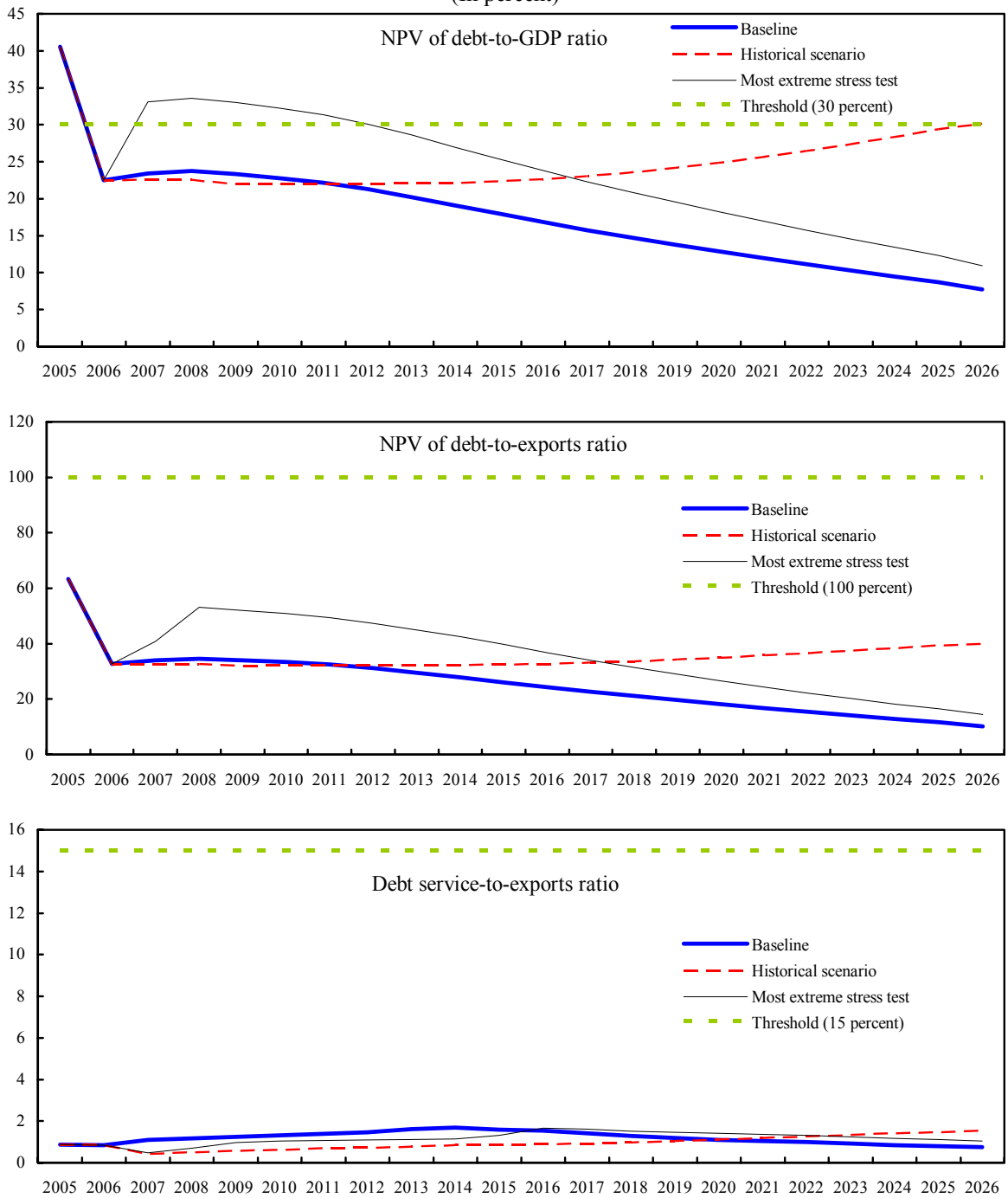
Table A4. Cambodia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2006-26
(In percent)

	Estimate	Projections						
	2006	2007	2008	2009	2010	2011	2016	2026
NPV of debt-to-GDP ratio								
Baseline	22	23	24	23	23	22	17	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	22	23	23	22	22	22	23	30
A2. New public sector loans on less favorable terms in 2007-26 2/	22	24	25	25	25	25	21	12
B. Bound Tests								
B1. Real GDP growth at baseline minus one standard deviation in 2007-08	22	24	25	24	24	23	18	8
B2. Export value growth at baseline minus one standard deviation in 2007-08 3/	22	26	32	31	30	29	22	9
B3. US dollar GDP deflator at baseline minus one standard deviation in 2007-08	22	24	25	25	24	24	18	8
B4. Net non-debt creating flows at baseline minus one standard deviation in 2007-08 4/	22	25	27	26	26	25	19	8
B5. Combination of B1-B4 using one-half standard deviation shocks	22	26	31	30	29	28	21	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	22	33	34	33	32	31	24	11
NPV of debt-to-exports ratio								
Baseline	33	34	34	34	33	33	24	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	33	33	33	32	32	32	33	40
A2. New public sector loans on less favorable terms in 2007-26 2/	33	35	37	37	37	37	30	16
B. Bound Tests								
B1. Real GDP growth at baseline minus one standard deviation in 2007-08	33	34	34	34	33	33	24	10
B2. Export value growth at baseline minus one standard deviation in 2007-08 3/	33	41	53	52	51	49	37	14
B3. US dollar GDP deflator at baseline minus one standard deviation in 2007-08	33	34	34	34	33	33	24	10
B4. Net non-debt creating flows at baseline minus one standard deviation in 2007-08 4/	33	36	39	39	38	37	27	11
B5. Combination of B1-B4 using one-half standard deviation shocks	33	38	45	44	43	42	31	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	33	34	34	34	33	33	24	10
Debt service ratio								
Baseline	0.8	1.1	1.2	1.2	1.3	1.4	1.5	0.7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	0.8	0.4	0.5	0.6	0.6	0.7	0.9	1.5
A2. New public sector loans on less favorable terms in 2007-26 2/	0.8	0.4	0.6	0.7	0.9	0.9	1.1	1.0
B. Bound Tests								
B1. Real GDP growth at baseline minus one standard deviation in 2007-08	0.8	0.4	0.5	0.6	0.7	0.7	0.8	0.7
B2. Export value growth at baseline minus one standard deviation in 2007-08 3/	0.8	0.5	0.7	1.0	1.0	1.1	1.7	1.0
B3. US dollar GDP deflator at baseline minus one standard deviation in 2007-08	0.8	0.4	0.5	0.6	0.7	0.7	0.8	0.7
B4. Net non-debt creating flows at baseline minus one standard deviation in 2007-08 4/	0.8	0.4	0.6	0.7	0.8	0.8	1.1	0.8
B5. Combination of B1-B4 using one-half standard deviation shocks	0.8	0.4	0.6	0.8	0.9	0.9	1.3	0.9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	0.8	0.4	0.5	0.6	0.7	0.7	0.8	0.7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	43	43	43	43	43	43	43	43

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure A2. Cambodia: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2005-2026
(In percent)



Source: Staff projections and simulations.
1/ Debt rescheduling with the Russian Federation and the United States is assumed in 2006.
2/ Most extreme stress test is test that yields highest ratio in 2015.

Figure A-3 : External Debt Indicators

Figure A3-1: NPV of External Debt to GDP (in percent)

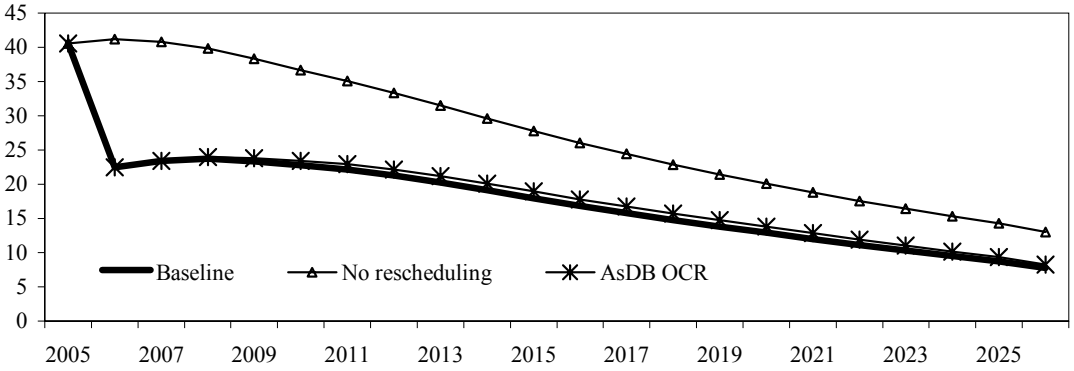


Figure A3-2: Debt Service to Exports (in percent)

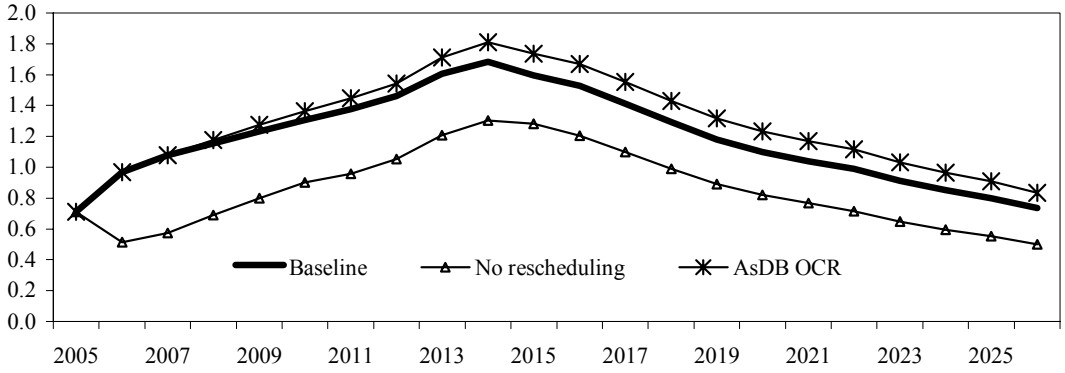
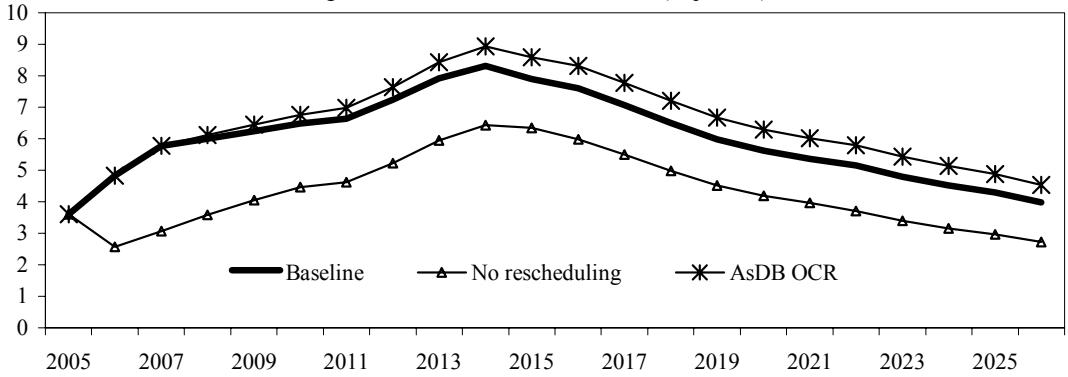


Figure A3-3: Debt Service to Revenue (in percent)



1/ Baseline scenario assumes that debt rescheduling with the U.S. and the Russian Federation will be implemented in 2006.
2/ No rescheduling scenario assumes that debt rescheduling with the U.S. and the Russian Federation will not be implemented.
3 AsDB OCR scenario assumes that the Asian Development Bank will provide non-concessional loan (Ordinary Capital Resources lending) US$20 million per year from 2008.
4/ Currently no debt service is paid to the U.S. and the Russian Federation. If the rescheduling agreement with the U.S. and the Russian Federation is not reached, Cambodia will pay less debt service under "No rescheduling" scenario comparing with baseline scenario and keep high debt outstanding level.